

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

**Machine Learning [ECE-4052]**

**Marks: 50**

**Duration: 180 mins.**

### PART-A

**Answer all the questions.**

- 1) List three machine learning algorithms for solving classification problems. (2)
- A)
- B) Explain why dimensionality reduction is useful in machine learning. (3)
- C) Given the following data with three samples and two features, compute the principal component vectors and the first principal components: (5)

$$X = \begin{bmatrix} 2 & 3 \\ 4 & 7 \\ 5 & 8 \end{bmatrix} \quad (5)$$

- 2) You are tasked with implementing LDA for a multi-class classification problem. The dataset has three classes, each with two features. Describe the steps involved in applying LDA to a multi-class classification problem. How does LDA handle multiple classes, and what modifications are necessary compared to the binary case? (5)
- A)

- B) Given the following dataset, using elementary geometry find the maximum margin hyperplane for the data. Verify the result by finding the same using the SVM algorithm. (5)

| Example | $x_1$ | $x_2$ | Class label |
|---------|-------|-------|-------------|
| 1       | 2     | 1     | -1          |
| 2       | 4     | 5     | +1          |
| 3       | 3     | 6     | +1          |

- 3) Design a two layer network of perceptrons to implement A XOR B. (5)
- A)
- B) (5)

Given the following data, use ANN with one hidden layer, appropriate initial weights and biases to compute the optimal values of the weights. Perform one iteration of the forward and phases of the backpropagation algorithm for each samples.

| Sample | Input<br>1 | Input<br>2 | Output target<br>1 | Output target<br>2 |
|--------|------------|------------|--------------------|--------------------|
| 1      | 1.20       | 2.30       | 0.53               | 0.76               |
| 2      | 0.23       | 0.37       | 1.17               | 2.09               |

4) Distinguish Joint probability vs conditional probability (2)

A) (2)

B) Explain how the choice of initialization can affect the convergence and final results of the K-means algorithm. (3)

C) Given the training data listed in the following table. Predict the class of the following new example using Naïve Bayes Classification.

| Sl. No | Refund | Marital Status | Taxable Income | Evade |
|--------|--------|----------------|----------------|-------|
| 1      | Yes    | Single         | 125K           | No    |
| 2      | No     | Married        | 100K           | No    |
| 3      | No     | Single         | 70K            | No    |
| 4      | Yes    | Married        | 120K           | NO    |
| 5      | No     | Divorced       | 95K            | Yes   |
| 6      | No     | Married        | 60K            | No    |
| 7      | Yes    | Divorced       | 220K           | No    |
| 8      | No     | Single         | 85K            | Yes   |
| 9      | No     | Married        | 75K            | No    |
| 10     | No     | Single         | 90K            | Yes   |

5) Explain how the discount factor (gamma) in reinforcement learning affects the agent's behavior. (2)

A) (2)

B) State the difference between a policy and a value function in reinforcement learning. How do they each guide an agent's decisions? (3)

C) Explain how decision trees can be used for both classification and regression tasks. What are the differences in terms of the output and the way the tree is constructed? (5)

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